Evaluation Methods

1 Overview on LLM-as-a-Judge*

Zheng et al. (2023) coined the expression "LLM-as-a-judge" to denote "a scalable and explainable way to approximate human preferences, which are otherwise very expensive to obtain". Within this framework, the concept of "Language-Model-as-an-Examiner" (Gu et al. 2024), referring to situations which require the use of "Likert scale scoring functions as an absolute evaluative measure", is particularly relevant.

The main challenge is to improve the evaluation performances of LLMs, and the literature distinguishes among three approaches:

- Design strategy of evaluation prompts
- Improvement strategy of LLMs' evaluation capabilities
- Optimization strategy of final evaluation results

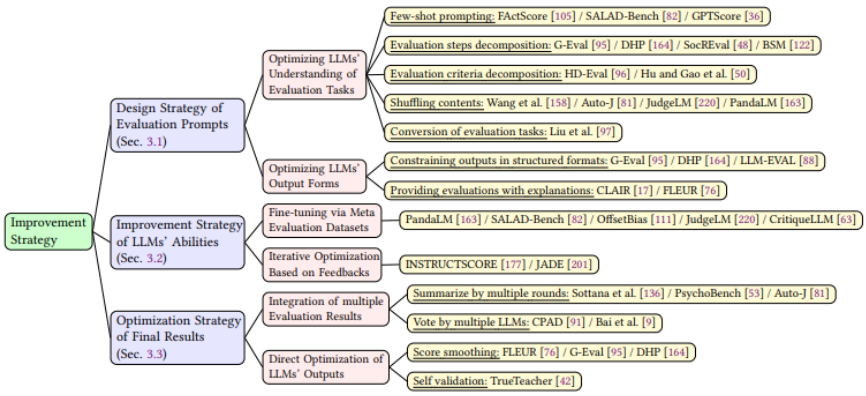

Figure 1: Summary of the main methods for the improvement of evaluation performances

The first approach focuses on the input provided to the LLM to guide the evaluation task. To improve the performance, it is fundamental to provide the LLMs prompts which give an optimized understanding of the evaluation task. The primary method in this regard is the so-called **few-shot prompting**, which involves "incorporating several high-quality evaluation examples into the evaluation prompts". Further methods to refine the evaluation task are

*This section is mainly based on Gu et al. (2024)

- **Decomposition of Evaluation Steps**, which entails "breaking down the entire evaluation tasks into smaller steps, providing detailed definitions and constraints for each small step in prompts"; this class includes methods related to the Chain-of-Thought

- **Decomposition of Evaluation Criteria**, which "involves breaking down the coarse evaluation criteria like Fluency into finer-grained sub-criteria like Grammar, Engaginess and Readability, and then generating overall scores based on these different dimensions"

Among the decomposition of evaluation criteria strategies, the **HD-eval** framework (Y. Liu et al. 2024) appears particularly interesting. The authors propose a three-stage process to achieve better alignment of LLM evaluations. The first step is represented by Hierarchical Criteria Decomposition, which "decompose an evaluation task into a hierarchy of evaluation criteria, each focusing on different valuation aspects with various granularity". The first layer of the decomposition is provided by human evaluators, while the following layers (further decomposition of the first one) are provided by the LLM. The following step, Human Preference-Guided Aggregation, estimates "how human experts value each decomposed criteria". Lastly, Attribution pruning "prune[s] the space of decomposition, focus[ing] on significant aspects, thus improving its fidelity".

The improvement of LLMs' evaluation capabilities can be instead pursued through **fine-tuning via meta evaluation dataset**, which involves collecting and constructing datasets specifically for the evaluation task. Another proposal regards the **iterative optimization based on feedback of evaluation results**, based on feedbacks provided either from a stronger model or directly from human evaluators' correction of the evaluation results.

Finally, optimization strategies of final results address "the inherent generation randomness within the black box of LLMs". The literature identifies mainly three proposals to tackle the issue:

- **Integration of multiple evaluation results**, obtained on the same content but with different hyperparameters; alternatively, the same content can be evaluated by multiple LLMs

- **Direct optimization of LLMs' output**, which involves "further processing the evaluation output to make it more reliable"; Y. Lee et al. (2024) proposed under this logic a score smoothing method

- **Conversion of evaluation tasks from pointwise evaluation to pairwise comparison**

2 Current trends

Schroeder and Wood-Doughty (2024) clarify some concepts related to the quality of LLMs evaluations. A key role is recognized to three distinct concepts:

- Reliability, referring to the "consistency of an LLM's output"
- Confidence, measuring "how certain [the LLMs] are about their responses"
- Calibration, measuring "how well this confidence aligns with the LLM's actual accuracy"

The authors focus on the idea of reliability and conduct an experiment. Their results indicate that "the ideal temperature for reliable LLM judgment is not universal; it varies depending on the specific LLM and the task". Furthermore, there could be a trade-off between reliability and performance[†].

Tian et al. (2024) examine the relation between prompting strategies and performance in automatic evaluation- They distinguish four prompting strategies

1. Few-shot-rubric, including rubric statement and examples[‡]
2. Few-shot-basic, including examples but not a rubric
3. Zero-shot-basic, including a rubric but not examples
4. Zero-shot-basic, including neither a rubric nor examples

The study reaches two major conclusions:

- "LLM evaluation performance is task-dependent", even though poorer performances appears to be linked to tasks with lower consensus among human graders
- "Effectiveness of contextual examples in the prompts templates"; in particular, The two zero-shot approaches are found to consistently underperform the other prompting strategies. However, for the remaining two, evidence is mixed[§]

The study suggests further exploration of methods related to chain-of-thought (CoT) prompting. CoT prompting involves providing the language model "a prompt that consists of triples: input, chain of thought, output" where the second element is represented by "a series of intermediate natural language reasoning steps that lead to

[†]Performance is here considered as the ranking in the Chatbot Arena(Chiang et al. 2024); in particular, "the ranking of LLM judgment reliability [...] inverts their Chatbot Arena performance ranking" (Schroeder and Wood-Doughty 2024)

[‡]"Each example includes the relevant component(s) from the learner-created artifact, human-grader's score, and their accompanying rationale text" (Tian et al. 2024)

[§]"The strategies that include rubric criteria tend to produce responses more consistent with the evaluation criteria. However, the rationale provided under the few-shot-rubric strategy [...] merely echoes the rubric statement without discussing specifics about the artifact" (Tian et al. 2024)

the final output" Wei et al. (2022).

This approach proved to be useful for math and logical reasoning problem, but its extension to text evaluation is not straightforward. G.-G. Lee et al. (2024) examine the use of CoT in the automatic scoring of student-written responses to science assignment[¶]. The evidences lead to the conclusion that "CoT is an efficient strategy for improving prompt engineering, particularly for the automatic scoring performance; yet CoT cannot work in isolation. CoT can be effective only when paired with the problem context and scoring rubrics".

A particular refinement of CoT approach is developed by Q. Zhang et al. (2025) as an efficient alternative to traditional majority voting and criteria expansion. This approach is called Crowd-based Comparative Evaluation (CCE), as it "first gathers a set of alternative responses to the task instruction, referred to as crowd responses, and then compare each candidate response against these crowd responses to derive multiple crowd judgements".

An interesting study (Kim and Oh 2025) investigates the use of LLMs to evaluate the creativity of texts. The study is particularly interesting as explains that "evaluating creative writing presents significant challenges, particularly due to the inherent ambiguity of creativity as a concept". The authors adopt an evaluation rubrics based on six attributes[‖], each of them assessed using an evaluation scale from 1 to 3. Comparing the results obtained by LLMs and human annotators, some conclusions emerge:

- Compared to human evaluators, LLMs achieve substantially higher level of Inter-Annotator Agreement, both across different temparatures and personas
- LLMs preserve this consistency even in situation which present a clearly biased or flawed valuation. As an example, LLMs did not recognize that some of the texts ended with an incomplete sentences

[¶]Note however that the assessment tasks involved simply or trinomial scoring

[‖]The attribute were the followings: fluency, flexibility, originality, elaboration, usefulness, specific creativity

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